

# hyper2kvm

## Web Dashboard — Product Overview

Browser-based VM migration management with real-time monitoring, multi-provider connectivity, and automated libvirt deployment

h2kweb v0.1.0 — Go + React + Tailwind + WebSocket

<http://185.165.240.5:5070>

# Architecture Overview

Full-stack Go backend with React frontend, integrated h2kvmctl CLI execution

Architecture Diagram

```
Browser → React Dashboard (Tailwind CSS, TanStack Query, Zustand)
  ↓ REST API + WebSocket
Go Backend (Chi v5 HTTP Router)
  ├── PAM Authentication (system login)
  ├── Provider Registry (vSphere, Azure, EC2)
  ├── Job Manager (in-memory queue)
  ├── WebSocket Hub (real-time broadcast)
  ├── File Browser API (server filesystem)
  ├── Host Info API (CPU, RAM, disk, libvirt)
  └── h2kvmctl Runner (subprocess + progress parsing)
      ↓ exec.Command + scanLinesOrCR
h2kvmctl (Python CLI) → Convert → Fix → Deploy → virsh define+start
```



## Go Backend

20MB static binary, Chi v5 router,  
WebSocket, PAM auth, systemd service



## React Frontend

Tailwind CSS (v9s design), TanStack  
Query, Zustand, Lucide icons, 87KB  
gzipped

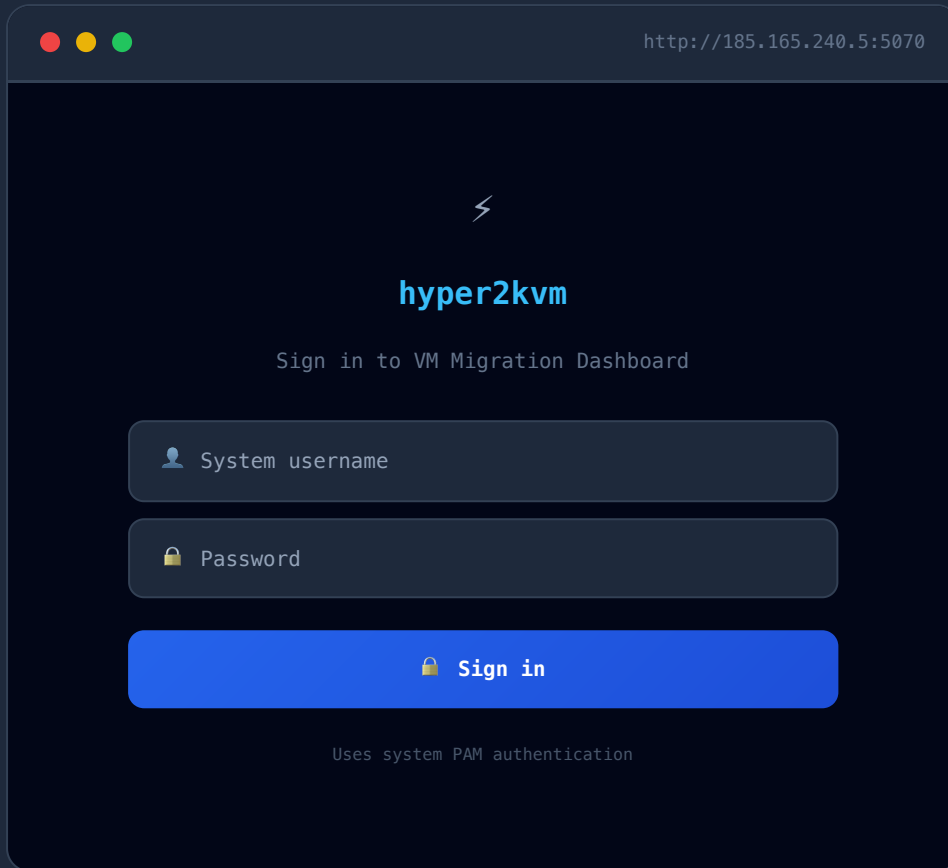


## h2kvmctl Integration

Subprocess execution, YAML config  
generation, live stdout parsing, progress  
tracking

# PAM Authentication

System-level authentication using Linux PAM — same credentials as SSH

A screenshot of a web browser window showing the login page for 'hyper2kvm'. The browser's address bar displays 'http://185.165.240.5:5070'. The page has a dark background. At the top center is a white lightning bolt icon. Below it, the text 'hyper2kvm' is displayed in a light blue font. Underneath that, it says 'Sign in to VM Migration Dashboard'. There are two input fields: the first is labeled 'System username' with a person icon, and the second is labeled 'Password' with a lock icon. Below these fields is a large blue button with a lock icon and the text 'Sign in'. At the bottom of the page, it says 'Uses system PAM authentication'.

## Security Features

- ✓ PAM authentication against system users
- ✓ HttpOnly session cookies (SameSite=Strict)
- ✓ 32-byte random hex session tokens
- ✓ In-memory session store (no external DB)
- ✓ API key fallback for CLI/scripts
- ✓ Public endpoints: /health, /auth/\* only
- ✓ All API routes require valid session
- ✓ Username character validation
- ✓ CORS middleware for browser access

## Auth Flow

```
POST /api/v1/auth/login
  → PAM verify (pam.StartFunc "login")
  → Create session (32-byte token)
  → Set-Cookie: h2kweb_session=<token>
```



# Dashboard — Real-time Overview

Live host metrics, migration pipeline status, provider health, and activity feed

8

CPU Cores

15.6 GB

Memory (43% used)

219 GB

Disk (90% used)

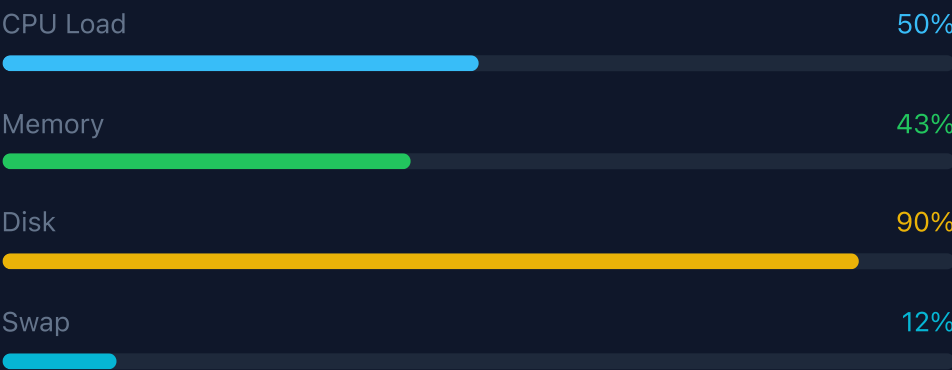
3

Libvirt VMs

## Host Information

|          |                                 |
|----------|---------------------------------|
| Hostname | NLDW3-4-31-12                   |
| OS       | AlmaLinux 9.7 (Moss Jungle Cat) |
| Kernel   | 5.14.0-611.41.1.el9_7.x86_64    |
| CPU      | Intel Xeon E3-1230 v6 @ 3.50GHz |
| QEMU     | 9.1.0                           |
| KVM      | ✓ Enabled                       |
| Uptime   | 3d 4h 59m                       |
| Load     | 4.04 / 4.16 / 4.42              |

## Resource Utilization



## Libvirt Domains

ubuntu2404

● running

centos10

● running

esxi-lab

● shut off

# Providers — Multi-Cloud Connectivity

Connect to VMware vSphere, Microsoft Azure, and AWS EC2 with dedicated login cards



## VMware vSphere

Connect to vCenter Server via govmomi SDK. Browse VMs, discover disks, export and migrate.

vCenter Host · Username · Password ·  
Datacenter



## Microsoft Azure

Connect via Azure CLI. List VMs by subscription and resource group, download managed disks.

Subscription ID · Resource Group



## AWS EC2

Connect via AWS CLI. Discover EC2 instances, export AMIs, migrate to KVM with full fixes.

Region · AWS Profile

## After Connecting — VM Browser

Provider → VM Browser

Search VMs...

|                                                   |                          |              |
|---------------------------------------------------|--------------------------|--------------|
| <input checked="" type="checkbox"/> web-server-01 | 4 vCPU · 8192MB · 120GB  | ● poweredOn  |
| <input checked="" type="checkbox"/> db-server-02  | 8 vCPU · 16384MB · 500GB | ● poweredOn  |
| <input type="checkbox"/> dev-sandbox              | 2 vCPU · 4096MB · 60GB   | ● poweredOff |
| <input type="checkbox"/> legacy-app               | 4 vCPU · 8192MB · 200GB  | ● poweredOn  |



# Migration Wizard

4 source types, server file browser, full YAML configuration, and YAML preview



### Local File

Browse server filesystem for  
VMDK, OVA, VHD



### VMware vSphere

Select from connected  
vCenter



### Microsoft Azure

Select from Azure  
subscription



### AWS EC2

Select EC2 instances

## Server File Browser

3 dirs 4 VM files

Show all files

vmware-exports/

|               |      |        |            |
|---------------|------|--------|------------|
| centos10.vmdk | VMDK | 2.2 GB |            |
| centos9.vmdk  | VMDK | 1.8 GB |            |
| fedora42.vmdk | VMDK | 4.6 GB | ✓ Selected |
| ubuntu.ova    | OVA  | 3.1 GB |            |

Selected: /root/demo/fedora42.vmdk VMDK

## Configuration Options (All YAML)

▼ Output

Output Directory · Format (qcow2/raw/vdi) · Filename

▼ Processing

☒ Flatten Snapshots · ☒ Compress Output · Level

▼ Guest Fixes

fstab Mode · ☒ Regen Initramfs · ☒ Update GRUB · ☒ Remove VMware Tools

▼ Libvirt / Deploy

☒ Emit XML · ☒ virsh define · ☒ Start VM · ☒ Keep Domain · Name/Memory/vCPUs

► Kubernetes

Deploy to KubeVirt · Namespace



# Job Monitor — Live Streaming

Real-time h2kvmctl output via WebSocket, deployment status, IP detection

## Live Output (WebSocket)

Terminal ● streaming 307 lines

```
1  ✓ Loading config: /tmp/h2kweb-xxx.yaml
2  ✓ Sanity: OK
3  ✓ Mode: local
4  ✓ Input VMDK: centos10.vmdk (2.20 GiB)
5  ✓ Quick probe: detected LVM on disk
6  ✓ Flattened: working-flattened.qcow2
7  ✓ VMCraft ready in 3.38s (1 drive)
8  ✓ LVM: cs-root, cs-home, cs-swap
9  ✓ Root: /dev/mapper/cs-root (xfs)
10 ✎ Running: dracut -f --add-drivers virtio...
11 ✓ Completed in 26.3s
12 ✓ grub2-mkconfig: Succeeded
13 ✓ Conversion complete: centos10.qcow2
14 ✓ Domain: centos10 → RUNNING
```

## Deployment Status



Libvirt

Domain

centos10

State

running



Network

● 192.168.122.45



SSH

## Job Card States

🕒 pending

Queued, waiting to start

🔄 running

Converting, with progress bar

✓ completed

Done, shows deployment status + IP

✗ failed

Error message displayed

# REST API Reference

12+ endpoints with PAM authentication, JSON responses, WebSocket streaming

| Method | Endpoint            | Description           |
|--------|---------------------|-----------------------|
| POST   | /auth/login         | PAM login             |
| POST   | /auth/logout        | Destroy session       |
| GET    | /auth/session       | Check session         |
| GET    | /health             | Health check          |
| GET    | /status             | Job stats + providers |
| GET    | /host/info          | CPU/RAM/disk/libvirt  |
| POST   | /providers          | Add provider          |
| GET    | /providers/{id}/vms | List VMs              |
| GET    | /browse?path=       | Browse filesystem     |
| POST   | /migrations/submit  | Start migration       |
| GET    | /jobs               | List all jobs         |
| GET    | /deploy/{vm}/ip     | VM IP detection       |

## Deployment

```
# Install (one command)
$ cd web && sudo make install

# Service management
$ systemctl status h2kweb
$ journalctl -u h2kweb -f

# Remote deploy
$ ./scripts/deploy-remote.sh 10.0.0.1 root

# Config: /etc/default/h2kweb
H2KWEB_ADDR=:5070
```

## Tech Stack

Go 1.23

Chi v5

React 18

Tailwind 3

TanStack Query 5

Zustand

WebSocket

PAM

govmomi

Lucide Icons

## Binary Size

|            |                       |
|------------|-----------------------|
| Go backend | 20 MB (static binary) |
|------------|-----------------------|

|    |     |                  |                 |               |
|----|-----|------------------|-----------------|---------------|
| WS | /ws | Real-time events | React frontend  | 88 KB gzipped |
|    |     |                  | CSS (Tailwind)  | 6 KB gzipped  |
|    |     |                  | systemd service | Built-in      |